Annex IV-Style Technical Documentation Summary: Positional-Encoding Ablation for Hourly Electricity-Demand Forecasting

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About this document

> This document uses the Annex IV structure of Regulation (EU) 2024/1689 as a documentation framework for the project. It does not represent a regulatory conformity assessment.

Sourcing. The legal text used here is Regulation (EU) 2024/1689 as published in the Official Journal on 12 July 2024 (ELI http://data.europa.eu/eli/reg/2024/1689/oj): Article 11, Annex IV, Article 6, Annex III, Article 9, Article 43 and recital 55. This is the version provided with the assignment, and later amendments, if any, are not reflected.

What Article 11 and Annex IV ask for. Article 11(1) says the technical documentation of a high-risk AI system is drawn up before the system is placed on the market or put into service, kept up to date, written to demonstrate compliance with the requirements of Chapter III, Section 2, and to give authorities and notified bodies the information to assess it "in a clear and comprehensive form". It must contain at least the elements of Annex IV, which has nine numbered points. This document follows those nine headings.

Classification and regulatory scope. Annex IV applies to high-risk systems, so a real file starts with Article 6. The only Annex III entry that could touch this system is point 2: AI systems "intended to be used as safety components in the management and operation of critical digital infrastructure, road traffic, or in the supply of water, gas, heating or electricity". Article 3(14) defines a safety component by its safety function, and recital 55 describes such components as systems used to directly protect the physical integrity of critical infrastructure or the health and safety of persons and property (its examples are water-pressure monitoring and fire-alarm control). My reading is that a demand forecaster used for planning is not obviously a safety component in that sense, so it would probably not fall under point 2. That is my reading of the text, not a legal opinion. Separately, Article 2(6) excludes AI systems developed and put into service solely for scientific research, and Article 2(8) excludes research, testing and development before a system is placed on the market. This study was never placed on the market, so the Regulation does not apply to it. This document uses the Annex IV structure to present the evidence in an auditor-readable way. If the system were high-risk under Annex III point 2, Article 43(2) would route it to conformity assessment by internal control (Annex VI), without a notified body.

Purpose and general description (Annex IV, point 1)

Intended purpose of the study. To decide whether the positional encoding of a small time-series transformer should be changed, judged on forecast accuracy and on runtime cost. The model reads 168 hours of demand for one meter and predicts the next 24 hours. The positional encoding is the only thing that differs between the three arms.

Arm	Positional encoding	Parameters
A	learned absolute embedding, 168 positions	162,520
B	fixed sinusoidal table added at the input	151,768
C	rotary (RoPE) on queries and keys in every layer	151,768

Not intended for. Use with real customer data, operational dispatch decisions, or any claim about transformers or RoPE in general. The data are synthetic and the model is small.

Hardware and software. Nine runs on one Tesla T4 in one Colab session, Python 3.13, `torch==2.11.0+cu130`, `numpy==2.1.3`, `PyYAML==6.0.3` (`requirements.lock.txt`). The repository runs the analysis and tests on CPU. There are no deployers, no user interface, no instructions for use (point 1(g), (h)) and no versioned release, because nothing was deployed. Point 1(f), photographs of a product, does not apply.

Development and design process (Annex IV, point 2(a) to (c), (e), (f))

Design choice and its rationale. A one-factor ablation. Shared backbone weights are initialised from the seed, the positional module from `seed + 10000` and training from `seed + 20000`, so the three arms start from bitwise-identical shared weights (checked by hash). Seeds 0, 1 and 2 are matched across arms, and differences are analysed per seed. Using matched seeds removes seed-to-seed variation in the shared weights from the comparison. It does not remove everything: arm A's three seeds differ by a standard deviation of 0.033 MAPE points.

Architecture. Three transformer encoder layers, model width 64, four heads, feed-forward width 256, dropout 0.1, mean pooling over positions and a linear head (output `1 + head(mean-pool)`).

Trade-offs and fixed choices.

- **Training length.** 8000 steps. Calibration looked for a step count where training loss improved by less than 2% between doublings. None of the tried grid values got there (4000 to 8000 still gave 5.12%), so the pre-declared fallback of 8000 was used. The models are not converged.
- **Determinism policy P1.** Deterministic algorithms, a fixed cuBLAS workspace, deterministic cuDNN and the math attention backend. P1 was chosen on feasibility alone (projected runtime against a three-hour ceiling), before seeing any accuracy result. A run repeated on one GPU matched bitwise. The cost is slower kernels, so latencies are not those of a tuned deployment.
- **No per-arm tuning, no early stopping, final-step weights only.**

Human oversight (point 2(e)) and pre-determined changes (point 2(f)). Point 2(e) asks for an assessment of the oversight measures a system needs under Article 14. I did not make that assessment, because there is no deployer and no decision the model's output feeds. The model has no pre-determined changes. What did exist was oversight of the experiment itself: the decision rule was frozen before the nine runs, and every stage needed explicit approval before the next began. Computational resources (point 2(c)): one Tesla T4, about 75 minutes in total for calibration, pilot and the nine main runs.

Data provenance and governance (Annex IV, point 2(d); Article 10)

Item	Detail
Source	Synthetic. A seeded generator (`gen-v1`, data seed 42) builds hourly demand for 16 meters over 3 years starting 2021-01-01.
Generator	A daily profile, a weekend factor, a latent temperature effect and AR(1) noise, with rare outlier hours (0.6 to 1.5 times) and short gaps of 1 to 6 hours. Gaps are filled with the last earlier value.
Fingerprint	SHA-256 `65d899e60a00802062ac5c9c0dc5202d8d5ed6e6a0d64cba593e8985fc53f695`, identical across machines.
Split	By time, no shuffling: 24 months train, 6 validation, 6 test. Windows are split by target time, start at midnight, stride 24 h.
Window counts	11,568 train, 2,912 validation, 2,928 test.
Leakage control	A leakage check runs with every data build (and in the test suite). Calibration used training loss only, and code gates block validation and test metrics for arms B and C in the calibration and pilot stages.
Labelling and cleaning	No labels: the target is the next 24 hours of the same series. Gaps are filled with the last earlier value. Outlier hours are part of the generated series and were not removed.
Personal data	None. Real data would bring Article 10 questions (representativeness, bias, personal data), which this exercise does not address.

Honest limit. Synthetic data can show whether the arms differ on this generator. It cannot show how they would behave on real load curves, whose structure the generator only imitates. The generator's weekly and daily patterns may also favour some positional schemes over others.

Training methodology (Annex IV, point 2(a), (b), (c))

AdamW, learning rate 0.001, weight decay 0.01, 100 warmup steps then linear decay, batch size 128, no gradient accumulation, fp32, normalised L1 loss. Main stage: three arms times seeds 0, 1 and 2 in one GPU session on the frozen commit, with the arm order rotated by seed (A,B,C / B,C,A / C,A,B) so that thermal or session drift does not line up with one arm. A calibration stage came first, followed by an arm-A noise pilot using the seed-900 calibration run plus additional seeds 901 and 902. Those results were not mixed into the main tables.

Pre-registration. Before the main runs, `PREREGISTRATION.md` and twelve other files were hashed (`results/freeze_record.json`). The runner refuses to start unless the record verifies. After the runs, `python src/freeze.py --verify` still reports that 13 files match.

Evaluation metrics and results (Annex IV, points 2(g) and 4)

Metric choice. Primary metric: validation MAPE (percent), because a percentage error is comparable across meters of different size and is the form a demand planner reads. Secondary: validation MAE in kW, test MAPE, training time, batch-1 inference latency (p50) and peak memory. MAPE is undefined near zero demand, so the generator enforces a positive floor; MAPE also weights errors at low demand more heavily than errors at peak demand, which may not match what matters operationally. Two seasonal baselines score 7.08 and 7.17.

Test logs (point 2(g)). Every run writes a JSON record with start and finish times, commit, GPU, metrics and hashes, and the results are in `results/runs/`. Annex IV asks for test reports dated and signed by the responsible persons. These records are dated but not signed.

Decision rule (frozen). For each comparison with arm A, d is the per-seed MAPE difference. A tentative call needs all three differences to share a sign, |mean| at least 0.10 points, |mean| at least 2 times the standard deviation of d, and every leave-one-out mean to stay beyond 0.10 in the same direction. These are exploratory screening criteria for three seeds, not significance tests. Three same-sign differences occur by chance 25% of the time.

Arm	Validation MAPE, seeds 0, 1, 2	Mean	Difference from A (per seed)	Mean difference	Frozen-rule verdict
A	4.634, 4.700, 4.666	4.667	n/a	n/a	reference
B	4.663, 4.632, 4.677	4.658	+0.028, -0.068, +0.011	-0.009	no difference detectable at this budget
C	4.745, 4.755, 4.772	4.757	+0.111, +0.055, +0.106	+0.090	no difference detectable at this budget

Reading it. C had higher error than A on all three seeds, but its mean difference (+0.090) is below the pre-registered threshold of 0.10, so the rule does not call it. B and A show mixed signs. Test MAPE moves the same way (A 4.20 to 4.26, B 4.21 to 4.23, C 4.28 to 4.31).

Cost under P1 on one T4. Training time 312.8 s (A), 312.9 s (B) and 343.4 s (C), so C was 9.8% slower in every matched pair. Batch-1 p50 latency 4.34, 4.32 and 4.69 ms. Peak training memory about 623 MiB in all arms. These describe this model, hardware, workload and deterministic configuration. They are not a general statement about RoPE.

Release recommendation (within the scope of this experiment). The evidence is insufficient to recommend changing the positional encoding for accuracy. Learned and sinusoidal encodings showed no detectable difference at the tested budget. RoPE added a measurable runtime cost without an accuracy improvement that met the rule.

Limitations, known risks and residual risk (Annex IV, points 3 and 5)

Risk or limit	Why it matters	What was done	Residual
Three seeds	The spread of three differences is a rough estimate, and a three-seed sign pattern can arise by chance.	Frozen, exploratory rule; verdicts worded as “no difference detectable at this budget”.	A real difference below about 0.1 points would not be reliably detected.
Noise underestimated	The arm A pilot SD (0.0073) was well below the main-run SD (0.0329).	Reported; delta was not lowered to fit the pilot.	Seed noise at this budget is larger than the pilot suggested.
Not converged	At 8000 steps the loss was still falling, and the ranking could change with longer training.	Stated as a fallback choice; extension not run.	Conclusions apply to this budget only.
Synthetic data	Real load data has structure, outliers and drift the generator lacks.	Provenance and hash documented.	No evidence about real data.
Single GPU and policy	Latency and memory depend on kernels; P1 is slower than defaults.	Environment pinned; hardware stated.	Cost figures do not transfer to other hardware or to default kernels.
Metric choice	MAPE over-weights low-demand hours.	MAE in kW and test MAPE also recorded.	An operational cost metric was not evaluated.
Bias and subgroup testing	A real file would test performance across customer groups and conditions.	Not applicable to synthetic meters.	Not assessed.
Cybersecurity and robustness	A real system needs Article 15-style checks.	Not assessed.	Not assessed.
Optional extension (seeds 3 and 4)	More seeds would tighten the estimate.	Not run; the three-seed results stay primary.	Open.

Monitoring, lifecycle, standards, declaration, post-market (Annex IV, points 3 and 6 to 9)

Annex IV section	Status in this exercise
Point 3: Monitoring, functioning and control	Capabilities, limitations and foreseeable unintended outcomes are in the limitations section. Run files record commit, GPU id, session, policy, config hash and data and weight hashes. No deployed system exists, so no operational logging or alerts.
Point 6: Changes through the lifecycle	Changes are logged in `EXPERIMENT_PLAN.md`, the amendment log in `PREREGISTRATION.md`, the git history and `session_manifest.json`. One deviation is recorded: the 8000-step calibration used commit `198aad8` rather than the pinned `c8ab0d2`, and a diff showed no run-executing code changed.
Point 7: Harmonised standards	None applied. Point 7 then asks for a description of the solutions adopted and other standards used; this document is that description. The crosswalk below maps the work to two frameworks, which is not a claim of conformity to any standard.
Point 8: EU declaration of conformity (Article 47)	**Not produced.**
Point 9: Post-market monitoring plan (Article 72(3))	**Not produced.** There is no market placement to monitor.

Crosswalk to NIST AI RMF and ISO/IEC 42001

This maps activities to the framework headings at topic level. It does not claim conformity, and I have not cited clause numbers.

NIST AI RMF function	What this project did	ISO/IEC 42001 theme (AI management system)
Govern	Staged approvals, a frozen pre-registration, a written plan and amendment log, defined roles (analyst approves each step).	Leadership, policy, roles, documented information
Map	Stated purpose and non-purposes, assumptions, data provenance, a classification caveat (About this document).	Context of the organisation, scope, AI system impact and risk identification
Measure	Pre-declared metric and decision rule, matched seeds, seed variation reported, cost measured, baselines.	Performance evaluation, internal review of results
Manage	Limitations table with residual risk, deviations logged, optional extension deferred, no deployment.	Risk treatment, continual improvement, change control

Where the project does not meet the framework's intent (no incident response, no deployer feedback, no organisational accountability beyond one analyst), the gap is real and is not covered by this table.